12.8.3 Routes through which microorganisms enter nervous system

- Skull or bone fractures
- Medical procedures
- Peripheral nerves
- Blood or lymph

12.8.4 Clinical Manifestations of Nervous System Infections

Some of the symptoms of nervous System infections are headache, fever, stiff neck, focal signs, seizures, confusion, weakness, hallucinations, stupor, coma, abnormal behavior and sleep disorder

12.8.5 Infections of Nervous System

- Meningitis is an inflammation of the meninges (membrane covering the brain). Meningitis is a diffuse infection caused by a variety of different agents.
- Encephalitis is defined as inflammation of the brain. Unlike an abscess, which is a localised area of bacterial or fungal growth, Encephalitis is usually due to viruses that produce more widespread intracellular infections.
- Brain abcess is a focus of purulent infection and is usually due to bacteria. Brain abscesses develop from either a contiguous focus of infection (such as the ears, the sinuses, or the teeth) or hematogenous spread from a distant focus (such as the lungs or heart, particularly with chronic purulent pulmonary disease, subacute bacterial endocarditis, or cyanotic congenital heart disease). In many cases the source is undetectable.

Etiological agents of Meningitis

This can be caused by a wide range of microorganisms and can be classified as pyogenic and non pyogenic meningitis. In pyogenic meningitis infiltration of pus cells (neutrophils) will be seen. In Non pyogenic or aseptic meningitis infiltration of lymphocytes may be seen. Diseases of nervous system are listed in Table 12.10.



Drugs cannot cross the blood brain barrier unless they are lipid soluble. Glucose and many amino acids are not lipid soluble, but they can cross the barrier through special transport systems. The lipid soluble antibiotic Chloramphenicol enters the brain readily. Penicillin is only slightly lipid soluble, but, if it is taken in very large doses, enough may cross the barrier to be effective. Inflammations of the brain tend to alter the blood brain barrier in such a way as to allow antibiotics to cross that would not be able to cross if there were no infection.

Antibodies found in the normal CNS are derived from the serum and are present at low levels compared to serum levels. There are a few phagocytic cells and complement is also largely excluded. CSF is especially vulnerable because it lacks many of the defenses found in the blood, such as phagocytic cells. It is not easy for the microorganisms to enter CNS but it hampers their clearance once it is penetrated.

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